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$$5^{2x} + 5^3 = 5^{x+2} + 5^{x+1}$$

$$\Rightarrow (5^x)^2 + 125 = (5^x \cdot 5^2) + (5^x \cdot 5^1)$$

$$\Rightarrow y^2 + 125 = (y \cdot 25) + (y \cdot 5)$$

$$\Rightarrow y^2 + 125 = 25y + 5y$$

$$\Rightarrow y^2 - 30y + 125 = 0$$

Discriminant:

$$\Delta = (-30)^2 - 4 \cdot 1 \cdot 125 = 900 - 500 = 400$$

$$\Rightarrow y = \frac{-(-30) \pm \sqrt{400}}{2 \cdot 1} = \frac{30 \pm 20}{2}$$

$$\Rightarrow y_1 = \frac{30 + 20}{2} = 25$$

$$\Rightarrow y_2 = \frac{30 - 20}{2} = 5$$

y terug substitueren:

$$\Rightarrow 5^x = 25 \Rightarrow x = \log_5(25) = 2$$

$$\Rightarrow 5^x = 5 \Rightarrow x = \log_5(5) = 1$$

## References